

NEL' KAN, Russian Federation

WMO: **311520**

Lat: **57.650N** Lon: **136.150E** Elev: **1044** StdP: **14.15** Time Zone: **10.00 (E10)** Period: **97-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB 99.6%	Heating DB 99%	Humidification 99.6% DP	99.6% HR	99.6% MCDB	Humidification 99% DP	99% HR	99% MCDB	Coldest Month 0.4% WS	0.4% MCDB	1% WS	1% MCDB	MCWS/PCWD to 99.6% DB MCWS	PCWD	WSF	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	12	-49.8	-45.8	-54.0	0.2	-49.1	-50.1	0.3	-45.5	9.1	-9.9	7.7	-18.3	1.1	180	N/A	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB 0.4% DB	0.4% MCWB	1% DB	1% MCWB	2% DB	2% MCWB	Evaporation WB/MCDB 0.4% WB	0.4% MCDB	1% WB	1% MCDB	2% WB	2% MCDB	MCWS/PCWD to 0.4% DB MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	21.6	85.4	65.9	81.3	63.5	77.8	62.2	68.2	81.8	66.0	78.0	63.8	74.8	2.9	180	(2)

	Dehumidification DP/MCDB and HR 0.4% DP	0.4% HR	0.4% MCDB	1% DP	1% HR	1% MCDB	2% DP	2% HR	2% MCDB	Enthalpy/MCDB 0.4% Enth	0.4% MCDB	1% Enth	1% MCDB	2% Enth	2% MCDB	Extreme Max WB	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	63.1	89.8	74.8	60.9	82.9	72.3	59.0	77.6	69.9	33.2	81.7	31.4	78.1	29.7	74.7	77.5	(3)

Extreme Annual Design Conditions

	Extreme Annual WS 1%	2.5%	5%		Extreme Annual Temperature Mean Min	Mean Max	Standard Deviation Min	Standard Deviation Max	n=5 years Min	n=5 years Max	n=10 years Min	n=10 years Max	n=20 years Min	n=20 years Max	n=50 years Min	n=50 years Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	9.1	7.6	6.9	DB	-53.6	90.5	5.2	3.4	-57.4	93.0	-60.4	95.0	-63.3	96.9	-67.1	99.3	(4)
(5)				WB	-53.7	71.8	5.1	3.1	-57.3	74.1	-60.3	75.9	-63.2	77.6	-66.9	79.8	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	21.0	-27.5	-19.1	4.3	27.9	47.1	59.5	65.1	60.2	45.0	23.4	-8.9	-27.7	(6)
(7)			DBStd	35.25	10.83	10.02	11.83	10.01	7.27	6.57	6.35	6.48	8.07	9.80	13.67	11.64	(7)
(8)			HDD50	11761	2402	1935	1417	663	144	6	0	5	191	823	1766	2409	(8)
(9)			HDD65	16191	2867	2355	1882	1113	554	187	77	177	601	1288	2216	2874	(9)
(10)			CDD50	1177	0	0	0	0	55	291	469	322	40	0	0	0	(10)
(11)			CDD65	132	0	0	0	0	0	21	82	29	0	0	0	0	(11)
(12)			CDH74	1722	0	0	0	0	23	358	1001	335	5	0	0	0	(12)
(13)			CDH80	480	0	0	0	0	2	68	333	77	0	0	0	0	(13)
(14)	Wind		WSAvg	3.0	2.9	2.7	3.0	3.1	3.7	3.1	2.7	2.9	2.9	3.0	2.9	2.9	(14)
(15)	Precipitation		PrecAvg	16.0	0.6	0.6	0.3	0.6	1.5	1.9	2.8	2.7	1.9	1.3	0.9	0.7	(15)
(16)			PrecMax	25.0	1.5	1.4	1.1	1.9	3.4	5.3	9.5	6.5	5.2	4.6	3.5	1.9	(16)
(17)			PrecMin	10.5	0.0	0.0	0.0	0.0	0.4	0.0	0.4	0.3	0.5	0.2	0.0	0.2	(17)
(18)			PrecStd	3.4	0.4	0.3	0.3	0.4	0.7	1.1	1.9	1.5	1.1	0.9	0.7	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		17.2	12.6	39.1	55.5	76.8	86.4	92.3	86.3	73.5	49.4	32.8	18.0	(19)
(20)			MCWB		14.7	10.3	31.1	43.5	57.2	63.2	70.1	67.9	58.5	41.2	28.3	15.9	(20)
(21)		2%	DB		-6.1	6.2	32.9	51.2	71.8	81.6	88.0	81.5	67.0	44.0	26.2	0.7	(21)
(22)			MCWB		-7.1	4.6	26.6	40.3	53.8	60.3	68.3	64.9	55.8	39.4	24.4	0.0	(22)
(23)		5%	DB		-10.1	1.8	28.1	47.5	67.4	78.1	83.7	77.6	61.9	40.2	17.5	-6.3	(23)
(24)			MCWB		-10.6	0.7	23.5	38.1	51.7	60.3	65.6	63.6	52.4	36.8	15.6	-6.9	(24)
(25)		10%	DB		-13.4	-3.8	23.8	43.6	62.5	74.8	79.6	73.4	57.6	36.3	10.7	-12.0	(25)
(26)			MCWB		-13.9	-4.8	20.1	35.4	49.1	58.5	63.6	61.3	50.0	32.8	9.2	-12.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		14.7	10.3	31.3	44.3	59.5	66.5	72.1	69.2	59.7	43.4	29.1	15.8	(27)
(28)			MCDB		17.1	12.1	39.0	53.8	74.0	78.6	88.2	82.5	70.6	47.5	32.3	17.9	(28)
(29)		2%	WB		-6.7	4.7	27.0	41.3	55.4	63.9	69.6	66.8	56.6	40.2	24.2	-0.1	(29)
(30)			MCDB		-6.1	6.1	32.6	50.1	69.1	76.0	84.9	79.0	66.0	43.6	26.2	0.8	(30)
(31)		5%	WB		-10.8	0.7	23.7	38.4	52.4	62.1	67.3	64.5	53.6	36.9	16.0	-6.9	(31)
(32)			MCDB		-10.3	1.8	28.0	46.8	65.4	75.0	80.6	75.5	61.1	40.2	17.5	-6.2	(32)
(33)		10%	WB		-13.8	-4.7	20.1	35.8	49.8	60.0	65.4	62.1	51.2	33.0	9.3	-12.5	(33)
(34)			MCDB		-13.3	-3.8	23.9	42.9	61.5	71.7	77.1	71.1	56.4	35.9	10.7	-12.0	(34)
(35)	Mean Daily Temperature Range		MDBR		11.8	18.9	22.9	21.0	21.0	22.7	21.6	20.3	17.3	11.5	13.6	10.2	(35)
(36)		5% DB	MCDBR		12.6	22.2	25.6	24.5	30.3	30.4	30.0	28.4	26.2	14.6	16.9	12.4	(36)
(37)			MCWBR		12.1	20.8	20.6	15.6	16.1	13.5	13.4	13.6	15.0	10.6	15.5	11.7	(37)
(38)		5% WB	MCDBR		12.5	22.2	24.8	23.1	28.5	25.0	27.1	25.6	23.5	13.2	16.6	12.4	(38)
(39)			MCWBR		12.0	20.9	19.8	15.2	16.5	12.4	13.9	13.4	14.4	10.2	15.3	11.7	(39)
(40)	Clear-Sky Solar Irradiance		taub		0.237	0.267	0.316	0.403	0.449	0.440	0.486	0.421	0.353	0.312	0.252	0.220	(40)
(41)			taud		2.056	2.091	2.011	1.858	2.012	2.181	2.054	2.248	2.388	2.246	2.062	2.000	(41)
(42)			Ebn at Noon		195	243	259	250	248	253	237	246	250	224	183	162	(42)
(43)			Edh at Noon		17	27	39	54	50	43	48	37	27	23	17	13	(43)
(44)	All-Sky Solar Radiation		RadAvg		239	589	1156	1702	1640	1809	1662	1302	853	452	291	156	(44)
(45)			RadStd		10	19	37	80	172	178	149	123	69	57	26	6	(45)

Historical Trends

		DBAvg	Heating 99% DB	Heating 99% DP	Cooling 1% DB	Cooling 1% WB	Cooling 1% DP	Degree-Days HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page